

SQL Grouping Sets

Knowledge Objectives

1. Enumerate the GROUPING SET clauses in SQL'99
2. Understand the different meaning of the NULL value

Application Objectives

1. Give the shortest query (using SQL'99 syntax) retrieving the necessary data to fill a statistical table

SQL Extension

Cross-tab view

	Time (Quarter)	Customer (City)				Product (Category)			
		Köln	Berlin	Lyon	Paris	Produce	Seafood	Condiments	Beverages
	Q1	21	10	18	35	35	14	23	17
	Q2	27	14	11	30	30	12	20	18
	Q3	26	12	35	32	32	10	33	18
	Q4	14	20	47	31	31	10	33	18



		Q1	Q2	Q3	Q4
Paris	Beverages	21	27	26	14
	Produce	10	14	12	20
	Condiments	18	11	35	47
	Seafood	35	30	32	31
Lyon	Beverages	12			
	Produce	20			
	Condiments	24			
	Seafood	33	14	12	10
Berlin	Beverages	33			
	Produce	25			
	Condiments	23			
	Seafood	25	23	20	33
Köln	Beverages	24			
	Produce	18			
	Condiments	28			
	Seafood	14	18	17	18

Marginal values

		Q1	Q2	Q3	Q4
Paris	Beverages	21	27	26	14
	Produce	10	14	12	20
	Condiments	18	11	35	47
	Seafood	35	30	32	31
Lyon	Beverages	12			
	Produce	20			
	Condiments	24			
	Seafood	33	14	12	10
Berlin	Beverages	33			
	Produce	25			
	Condiments	23			
	Seafood	25	23	20	33
Köln	Beverages	24			
	Produce	18			
	Condiments	28			
	Seafood	14	18	17	18



		Q1	Q2	Q3	Q4	Total
Paris	Beverages	21	27	26	14	88
	Produce	10	14	12	20	56
	Condiments	18	11	35	47	111
	Seafood	35	30	32	31	128
	Total	84	82	105	112	383
Lyon	Beverages	12				
	Produce	20				
	Condiments	24				
	Seafood	33	14	12	10	69
	Total	89				
Berlin	Beverages	33				
	Produce	25				
	Condiments	23				
	Seafood	25	23	20	33	101
	Total	106				
Köln	Beverages	24				
	Produce	18				
	Condiments	28				
	Seafood	14	18	17	18	67
	Total	84				
Total		363				

Cross-tab example

Sales	Catalonia		
	January23	February23	All
Ballpoint	275827	290918	566745
Rubber	784172	918012	1702184
All	1059999	1208930	2268929

SQL-92: Multidimensional query

Sales	Catalonia		
	January23	February23	All
Ballpoint	275827	290918	566745
Rubber	784172	918012	1702184
All	1059999	1208930	2268929

```
SELECT d1.itemName, d2.region, d3.month, SUM(f.items)
FROM Sales f, Product d1, Place d2, Time d3
WHERE f.IDProduct=d1.ID AND f.IDPlace=d2.ID
      AND f.IDTime=d3.ID
      AND d1.itemName IN ('Ballpoint','Rubber')
      AND d2.region='Catalonia'
      AND d3.month IN ('January23','February23')
GROUP BY d1.itemName, d2.region, d3.month
```

```
UNION
SELECT d1.itemName, d2.region, 'All', SUM(f.items)
FROM Sales f, Product d1, Place d2, Time d3
WHERE f.IDProduct=d1.ID AND f.IDPlace=d2.ID
      AND f.IDTime=d3.ID
      AND d1.itemName IN ('Ballpoint','Rubber')
      AND d2.region='Catalonia'
      AND d3.month IN ('January23','February23')
GROUP BY d1.itemName, d2.region
```

UNION

```
SELECT 'All', d2.region, d3.month, SUM(f.items)
FROM Sales f, Product d1, Place d2, Time d3
WHERE f.IDProduct=d1.ID AND f.IDPlace=d2.ID
      AND f.IDTime=d3.ID
      AND d1.itemName IN ('Ballpoint','Rubber')
      AND d2.region='Catalonia'
      AND d3.month IN ('January23','February23')
GROUP BY d2.region, d3.month
```

UNION

```
SELECT 'All', d2.region, 'All', SUM(f.items)
FROM Sales f, Product d1, Place d2, Time d3
WHERE f.IDProduct=d1.ID AND f.IDPlace=d2.ID
      AND f.IDTime=d3.ID
      AND d1.itemName IN ('Ballpoint','Rubber')
      AND d2.region='Catalonia'
      AND d3.month IN ('January23','February23')
GROUP BY d2.region
ORDER BY d1.itemName, d2.region, d3.month;
```

SQL-99: GROUPING SETS (I)

Sales	Catalonia		
	January23	February23	All
Ballpoint	275827	290918	566745
Rubber	784172	918012	1702184
All	1059999	1208930	2268929

```
SELECT d1.itemName, d2.region, d3.month, SUM(f.items)
FROM Sales f, Product d1, Place d2, Time d3
WHERE f.IDProduct=d1.ID AND f.IDPlace=d2.ID
      AND f.IDTime=d3.ID
      AND d1.itemName IN ('Ballpoint','Rubber')
      AND d2.region='Catalonia'
      AND d3.month IN ('January23','February23')
```

```
GROUP BY GROUPING SETS (
    (d1.itemName, d2.region, d3.month),
    (d1.itemName, d2.region),
    (d2.region, d3.month),
    (d2.region))
```

```
ORDER BY d1.itemName, d2.region, d3.month;
```

SQL-99: GROUPING SETS (II)

```
SELECT d1.itemName, d2.region, d3.month, SUM(f.items)
FROM Sales f, Product d1, Place d2, Time d3
WHERE f.IDProduct=d1.ID AND f.IDPlace=d2.ID
      AND f.IDTime=d3.ID
      AND d1.itemName IN ('Ballpoint','Rubber')
      AND d2.region='Catalonia'
      AND d3.month IN ('January23','February23')
GROUP BY GROUPING SETS ((d1.itemName, d2.region, d3.month),
                        (d1.itemName, d2.region),
                        (d2.region, d3.month),
                        (d2.region))
ORDER BY d1.itemName, d2.region, d3.month;
```

itemName	region	month	items
Ballpoint	Catalonia	January23	275827
Ballpoint	Catalonia	February23	290918
Ballpoint	Catalonia	NULL	566745
Rubber	Catalonia	January23	784172
Rubber	Catalonia	February23	918012
Rubber	Catalonia	NULL	1702184
NULL	Catalonia	January23	1059999
NULL	Catalonia	February23	1208930
NULL	Catalonia	NULL	2268929

SQL-99: Meaning of null value

```
SELECT
  CASE WHEN GROUPING(d1.itemName)=1 THEN 'AllBallpoint&Rubber'
        ELSE d1.itemName,
  d2.region,
  CASE WHEN GROUPING(d3.month)=1 THEN 'AllJanuary&February'
        ELSE d3.month,
  SUM(f.items)
FROM Sales f, Product d1, Place d2, Time d3
```

...

itemName	region	month	items
Ballpoint	Catalonia	January23	275827
Ballpoint	Catalonia	February23	290918
Ballpoint	Catalonia	AllJanuary&February	566745
Rubber	Catalonia	January23	784172
Rubber	Catalonia	February23	918012
Rubber	Catalonia	AllJanuary&February	1702184
AllBallpoint&Rubber	Catalonia	January23	1059999
AllBallpoint&Rubber	Catalonia	February23	1208930
AllBallpoint&Rubber	Catalonia	AllJanuary&February	2268929

SQL-99: ROLLUP (I)

GROUP BY ROLLUP ($a_1, \dots, a_n$)
equivalent to
GROUP BY GROUPING SETS (($a_1, \dots, a_{n-1}, a_n$),
($a_1, \dots, a_{n-1}$),
...,
(a_1),
($$))

SQL-99: ROLLUP (II)

```
SELECT d1.itemName, d2.region, d3.month, SUM(f.items)
FROM Sales f, Product d1, Place d2, Time d3
WHERE f.IDProduct=d1.ID AND f.IDPlace=d2.ID
      AND f.IDTime=d3.ID
      AND d1.itemName IN ('Ballpoint','Rubber')
      AND d2.region='Catalonia'
      AND d3.month IN ('January23','February23')
GROUP BY ROLLUP(d2.region, d1.itemName, d3.month)
ORDER BY d2.region, d3.month, d1.itemName;
```

itemName	region	month	items
Ballpoint	Catalonia	January23	275827
Rubber	Catalonia	January23	784172
Ballpoint	Catalonia	February23	290918
Rubber	Catalonia	February23	918012
Ballpoint	Catalonia	NULL	566745
Rubber	Catalonia	NULL	1702184
NULL	Catalonia	NULL	2268929
NULL	NULL	NULL	2268929

SQL-99: ROLLUP (III)

```
SELECT d1.itemName, d2.region, d3.month, SUM(f.items)
FROM Sales f, Product d1, Place d2, Time d3
WHERE f.IDProduct=d1.ID AND f.IDPlace=d2.ID
      AND f.IDTime=d3.ID
      AND d1.itemName IN ('Ballpoint','Rubber')
      AND d2.region='Catalonia'
      AND d3.month IN ('January23','February23')
GROUP BY d2.region, ROLLUP(d1.itemName, d3.month)
ORDER BY d2.region, d3.month, d1.itemName;
```

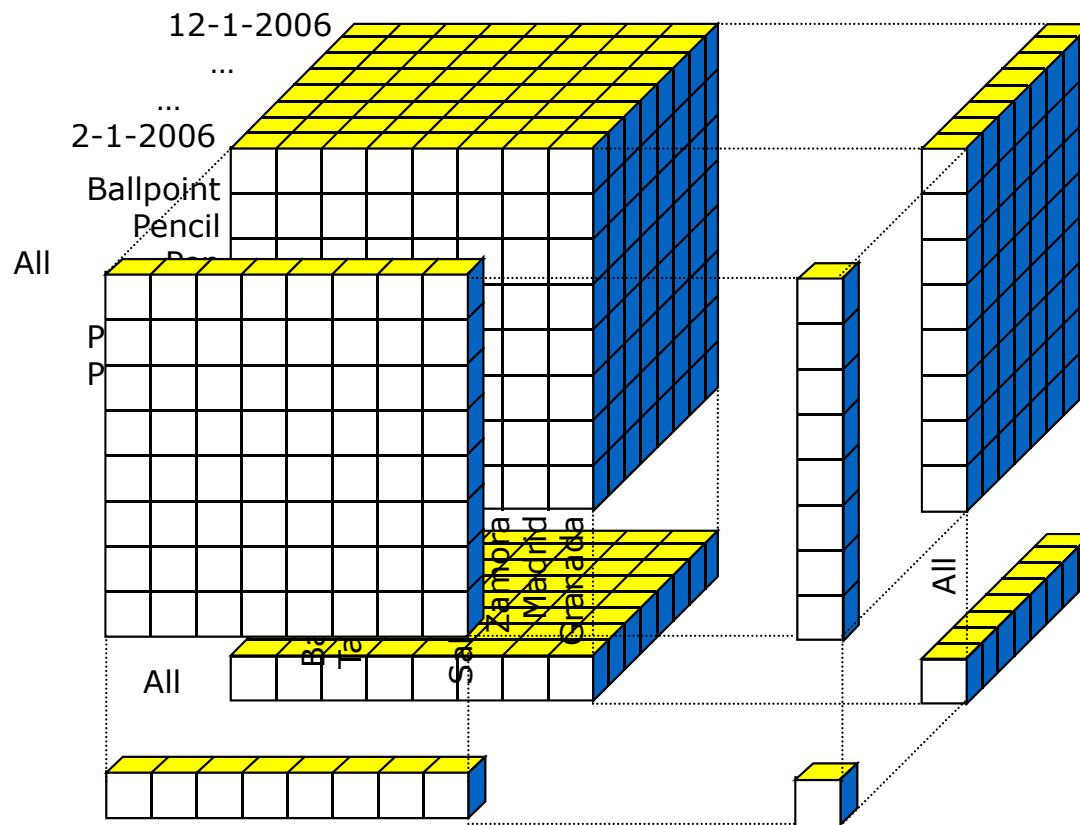
itemName	region	Month	items
Ballpoint	Catalonia	January23	275827
Rubber	Catalonia	January23	784172
Ballpoint	Catalonia	February23	290918
Rubber	Catalonia	February23	918012
Ballpoint	Catalonia	NULL	566745
Rubber	Catalonia	NULL	1702184
NULL	Catalonia	NULL	2268929

SQL-99: CUBE (I)

```
SELECT d1.itemName, d2.region, d3.month, SUM(f.items)
FROM Sales f, Product d1, Place d2, Time d3
WHERE f.IDProduct=d1.ID AND f.IDLugar=d2.ID
      AND f.IDTime=d3.ID
      AND d1.itemName IN ('Ballpoint','Rubber')
      AND d2.region='Catalonia'
      AND d3.month IN ('January23','February23')
GROUP BY d2.region, CUBE(d1.itemName, d3.month)
ORDER BY d1.itemName, d2.region, d3.month;
```

itemName	region	month	items
Ballpoint	Catalonia	January23	275827
Ballpoint	Catalonia	February23	290918
Ballpoint	Catalonia	NULL	566745
Rubber	Catalonia	January23	784172
Rubber	Catalonia	February23	918012
Rubber	Catalonia	NULL	1702184
NULL	Catalonia	January23	1059999
NULL	Catalonia	February23	1208930
NULL	Catalonia	NULL	2268929

SQL-99: CUBE (II)



SQL-99: CUBE (III)

GROUP BY CUBE (a,b,c)
equivalent to
GROUP BY GROUPING SETS ((a,b,c),
 (a,b),
 (a,c),
 (b,c),
 (a),
 (b),
 (c),
 ());

SQL-99: Combinations

GROUP BY CUBE(a,b), ROLLUP(c,d)
equivalent to
GROUP BY GROUPING SETS ((a,b,c,d),
 (a,b,c),
 (a,b),
 (a,c,d),
 (a,c),
 (a),
 (b,c,d),
 (b,c),
 (b),
 (c,d),
 (c),
 ())

Closing

Summary

- SQL-99
 - GROUPING SETS
 - ROLLUP
 - CUBE

References

- M. Golfarelli and S. Rizzi. *Data Warehouse Design*. McGraw-Hill, 2009
- A. Vaisman and E. Zimányi. *Data Warehouse Systems*. Springer, 2014